LOAD CALCULATIONS

1. DEAD LOAD

SELFWEIGHT FACTOR = 1

Dead load for Poly sheet assumed = 5 Kg/m^2 i.e. 0.05 kN/m^2

2. LIVE LOAD

Live load assumed = 0.75 kN/m^2

3. WIND LOAD

For wind load calculation, following IS code is referred: - "**IS 875 Part III**"

DESIGN WIND PRESSURE CALCULATION				
Sr no:	**Description**	**Value**	**Unit**	**Remarks**
1	V_b Basic Wind Speed	44	m/s	As per Apendix A (IS-875 Part 3)
2	K1 (risk coefficient)	1		For Design life span of 50 yrs. As per IS 875 Part III (table-1)
3	K2 (Terrain roughness and height factor)	1		As per IS 875 Part III (table-2), Cat-2, Ht 33M
4	K3 (topography factor)	1		Clause 6.3.3 IS 875 part III
5	K4 (Importance factor for Cyclonic region.)	1		Clause 6.3.4 IS 875 part III
6	K_a (Area Averaging factor)	0.80		As per IS 875 Part III (table-4)
7	K_d (Wind Directionality Factor)	0.9		Clause 7.2.1 IS 875 part III
8	K_c (Combination Factor)	0.9		Clause 7.3.3.13 IS 875 part III
	Vz=	Vb*K1*K2*K3*K4		
	Vz	**44**	**m/s**	
	Pz =	0.91	kN/m2	Pz =0.6(Vz)^2
	Pd1 =	0.59	kN/m2	Pd1=Pz*Ka*Kd*Kc
	or Pd2 =	0.64	kN/m2	Pd2=0.7*Pz
	Hence Use pd =	0.64	kN/m2	Max of Pd1 and Pd2
	Coefficeint for Pressure Case=	0.6		
	Coefficeint for Suction Case=	0.3		
	Span =	1		
	UDL for Pressure case =	0.4	kN/m	
	UDL for Suction case =	0.2	kN/m	